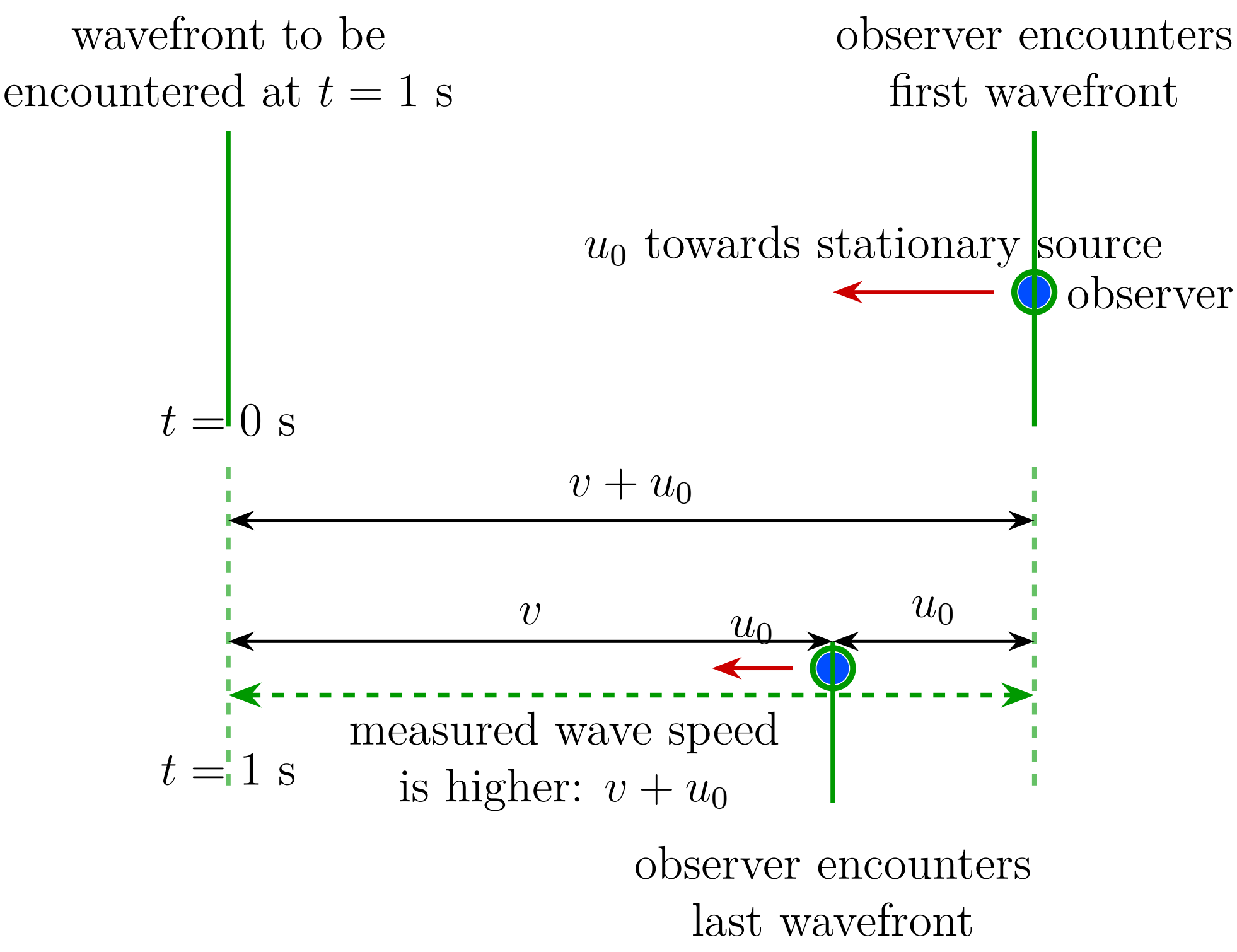
wavefront to be encountered at $t = 1$ s
observer encounters first wavefront
u_0 towards stationary source
observer
$t = 0$ s
$v + u_0$
v
u_0
u_0
measured wave speed is higher: $v + u_0$
$t = 1$ s
observer encounters last wavefront